Input: $x \in \{0,1\}^n$, $\rho \in \{0,\star\}^n$

Output: $a \in \{0,1\}$

$v \leftarrow$ root of T

while v is not a leaf **do**

  if $\operatorname{rank}(Q^{\prec v}_{S(\rho,\star)}) = \operatorname{rank}(Q^{\prec v}_{S(\rho,\star)}) + 1$ **then**

    Query $b^v \leftarrow \langle x, Q^v \rangle$

  else

    Infer $b^v \leftarrow \langle x, Q^v \rangle$ from the fact that $(Q^{\prec v})^T x = b^{\prec v}$ and $x_j = 0$ for all $\rho_j = 0$

  end if

  Move $v \leftarrow \operatorname{child}(v, b^v)$.

end while

return $L(v)$